2. Random Variables (Part I)

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MTH345: Deep Generative Models: A Statistical Perspective

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Lecture Topics

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- This lecture briefly reviews random variables, univariate and multivariate distributions, and the expectations of their functions.
- Useful materials include:
  - Mathematics for Machine Learning [MML] (Deisenroth et al., 2020)
  - A First Course in Probability [FCP] (Ross, 2020)
  - Mathematical Statistics and Data Analysis [MSDA] (Rice, 2007)
- See Chapter 6 in [MML] for details. We present content mostly following the textbooks, but also provide substantial unique and advanced material.
- Selected examples and text excerpts are reproduced from the materials for educational use only under fair use.

Outline

Random Variables

- With the probability, we can describe probabilistic models of real-world phenomena involving randomness.
- In the two-coin toss example, suppose we are interested in the number of heads. Before the experiment, we do not know which $s \in S = \{HH, HT, TH, TT\}$ will occur, so the result is random (the probability describes this uncertainty).
- However, once the random experiment yields a specific s (e.g., HH), the number of heads is deterministically fixed (e.g., 2).
- <u>**Random Variable**</u>: A *random variable* (r.v.) is a function that maps each possible outcome in S to a quantity of interest in the *target space* $\mathcal{T}$.
- For example, $X : S = \{HH, HT, TH, TT\} \to \mathcal{T} = \{0, 1, 2\}$ defined by:

$$X(HH) = 2, \quad X(HT) = 1, \quad X(TH) = 1, \quad X(TT) = 0 \tag{1}$$

is the r.v. representing the number of heads.

Random Variables

- Although the probability $P : \mathcal{A} \to [0, 1]$ and the r.v. $X : S \to \mathcal{T}$ have different domains, they are linked via the inverse image.
- For any subset $T \subseteq \mathcal{T}$, it is common to introduce the following notation:

$$P(X \in T) = P(\{s : X(s) \in T\}) = P(X^{-1}(T)). \tag{2}$$

- Formally, T must be a measurable set, meaning that $X^{-1}(T) \in \mathcal{A}$ so that its probability is defined under P.

Cumulative Distribution Function

- We first define distributions of univariate r.v.s ($X : S \to \mathbb{R}$), and then extend these concepts to multivariate ones (or *random vectors* $\boldsymbol{X} : S \to \mathbb{R}^d$).
- **(Cumulative) Distribution Function**: For a r.v. X,

$$F_X(x) = P(X \leq x), \quad -\infty < x < \infty \tag{3}$$

  is called the *cumulative distribution function (CDF)* or the *distribution function*.
- From now on, we omit the subscript denoting the random variable when the context is clear.
- In the two-coin toss example, where X represents the number of heads, the CDF of X is given by:

$$F(x) = \begin{cases} 0 & x < 0, \\ 1/4 & 0 \leq x < 1, \\ 3/4 & 1 \leq x < 2, \\ 1 & 2 \leq x. \end{cases} \tag{4}$$

Cumulative Distribution Function

- The CDF F satisfies the following properties:
  1. F is a non-decreasing function.
  2. $F(\infty) := \lim_{x \to \infty} F(x) = 1$.
  3. $F(-\infty) := \lim_{x \to -\infty} F(x) = 0$.
  4. F is right-continuous.

  We denote the left limit at x by $F(x-)$, which satisfies $F(x-) = P(X < x)$.
- Conversely, for any function F satisfying the above four properties, there exists a probability P and a r.v. X such that $F_X = F$.

Discrete, Continuous, and Mixed Random Variables

- **Types of Random Variables**:
  - X is *discrete* if F is a step function (piecewise constant).
  - X is *continuous* if F is a continuous function.
  - X is *mixed* if F is neither.

  In this course, for continuous r.v.s, we assume that F is differentiable, except possibly at countably many points.
- The target space $\mathcal{T}$ is countable when X is discrete, and uncountable otherwise.

Discrete, Continuous, and Mixed Random Variables

- For a discrete r.v., it is convenient to define a function to describe the size of the jumps in its CDF.
- **Probability Mass Function**: For a discrete r.v. X, the function p_X defined by

$$p_X(x) := P(X = x) = P(\{s : X(s) = x\}) \tag{5}$$

is called the *probability mass function* (PMF).

- The PMF is related to the CDF by the jump size:

$$p(x) = P(X \leq x) - P(X < x) = F(x) - F(x-). \tag{6}$$

- For example, when X represents the number of heads in the two-coin toss,

$$p(1) = P(\{HT, TH\}) = 1/2. \tag{7}$$

Discrete, Continuous, and Mixed Random Variables

- Similarly, for a continuous r.v., it is convenient to define a function to describe the infinitesimal rate of increase of the CDF.
- **Probability Density Function**: For a continuous r.v. X, a function f_X satisfying the following conditions is called the *probability density function* (PDF):
  1. $f_X(x) \geq 0$ for all $x \in \mathbb{R}$.
  2. $P(a \leq X \leq b) = \int_a^b f_X(x)dx$ for any $a \leq b$.[1]

  Note that $P(X = a) = \int_a^a f_X(x)dx = 0$ for any a.

[1] This course uses f_X to distinguish density from probability, but it is also common to use p_X to denote the PDF.

Discrete, Continuous, and Mixed Random Variables

- The PDF is related to the CDF by its derivative:

$$f(x) = \frac{d}{dx}F(x) = \lim_{\epsilon \to 0} \frac{F(x+\epsilon) - F(x)}{\epsilon} \tag{8}$$

(defined at points where F is differentiable). Consequently, for a small $\epsilon > 0$,

$$P(x - \epsilon/2 \leq X \leq x + \epsilon/2) \approx f(x)\epsilon. \tag{9}$$

- The probability of an interval is given by the difference in CDF values:

$$P(a < X \leq b) = F(b) - F(a). \tag{10}$$

Means and Variances

- **Expectation (Mean)**: The *expectation (mean)* of a discrete random variable X with probability mass function p_X is:

$$\mu_X = \mathrm{E}[X] = \sum_x x p_X(x). \tag{11}$$

Correspondingly, the expectation of a continuous random variable X with probability density function f_X is:

$$\mu_X = \mathrm{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\,dx. \tag{12}$$

- **Law of the Unconscious Statistician**: If X is a discrete random variable, then for any real-valued function g, $\mathrm{E}[g(X)] = \sum_x g(x) p_X(x)$. Similarly, for a continuous random variable X, $\mathrm{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\,dx$.[2]

[2] See [FCP] Proposition 4.1 for details.

Means and Variances

- **Variance**: The *variance* of X is defined as:

$$\sigma_X^2 = \mathrm{Var}[X] = \mathrm{E}[(X - \mathrm{E}[X])^2]. \tag{13}$$

That is, $\mathrm{Var}[X]$ is the expected squared Euclidean distance between the possible values of X and its mean.

- Important properties of means and variances include:
  - $\mathrm{E}[aX + b] = a\,\mathrm{E}[X] + b$
  - $\mathrm{Var}[X] = \mathrm{E}[X^2] - \big(\mathrm{E}[X]\big)^2$
  - $\mathrm{Var}[aX + b] = a^2\,\mathrm{Var}[X]$
  - $\mathrm{Var}[X] = 0$ if and only if X is deterministic, i.e., $P(X = \mathrm{E}[X]) = 1$
- Using the mean and variance, the standardized random variable Z is defined as:

$$Z = \frac{X - \mathrm{E}[X]}{\sqrt{\mathrm{Var}[X]}} \tag{14}$$

where we assume $\mathrm{Var}[X] > 0$.

Outline

Univariate Discrete Random Variables

- Common univariate discrete distributions include the Bernoulli and binomial distributions.
- **Bernoulli Random Variable**: A random variable X is said to follow the *Bernoulli distribution*, denoted as $X \sim \text{Ber}(\mu)$ where $0 \leq \mu \leq 1$, if its PMF is given by:

$$p_X(x;\mu) = \begin{cases} \mu & \text{if } x = 1, \\ 1-\mu & \text{if } x = 0, \\ 0 & \text{otherwise.} \end{cases} \tag{15}$$

- The PMF of $X \sim \text{Ber}(\mu)$ can be summarized in the following table:

x	0	1
$p_X(x;\mu)$	$1-\mu$	μ

- The mean and variance of X are $\text{E}[X] = \mu$ and $\text{Var}[X] = \mu(1-\mu)$.

Univariate Discrete Random Variables

- **Binomial Random Variable**: A random variable X is said to follow the *binomial distribution*, denoted as $X \sim \mathrm{Bin}(n, \mu)$, where $n \in \mathbb{N}$ and $0 \leq \mu \leq 1$, if its PMF is given by:

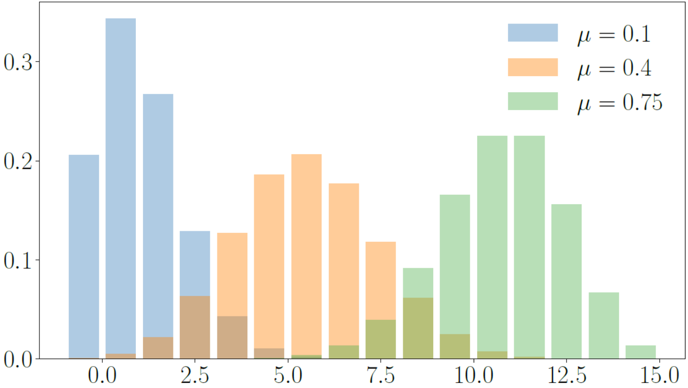
$$p_X(x; n, \mu) = \begin{cases} \binom{n}{x} \mu^x (1-\mu)^{n-x} & \text{if } x \in \{0, 1, \ldots, n\}, \\ 0 & \text{otherwise.} \end{cases} \tag{16}$$

  Here, $\binom{n}{x}$ denotes the binomial coefficient, defined as $\frac{n!}{x!(n-x)!}$, which represents the number of distinct ways to choose x objects from n distinguishable objects.
- The PMF of $X \sim \mathrm{Bin}(n, \mu)$ can be summarized in the following table:

x	0	1	$\ldots$	n
$p_X(x; n, \mu)$	$(1-\mu)^n$	$n\mu(1-\mu)^{n-1}$	$\ldots$	μ^n

- The mean and variance of X are $\mathrm{E}[X] = n\mu$ and $\mathrm{Var}[X] = n\mu(1-\mu)$.

Univariate Discrete Random Variables



[MML] Fig. 6.10: The PMFs of $\mathrm{Bin}(n, \mu)$ when $(n, \mu) = (15, 0.1)$, $(15, 0.4)$, and $(15, 0.75)$.

Univariate Continuous Random Variables

- Common univariate continuous distributions include the uniform and Gaussian distributions.
- **Uniform Random Variable**: A random variable X is said to follow the *uniform distribution* over $[a, b]$, denoted by $X \sim \mathcal{U}(a, b)$, if its PDF is given by:

$$f_X(x; a, b) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b, \\ 0 & \text{otherwise.} \end{cases} \tag{17}$$

- The mean and variance of $X \sim \mathcal{U}(a, b)$ are given by:

$$\mathrm{E}[X] = \frac{a+b}{2} \quad \text{and} \quad \mathrm{Var}[X] = \frac{(b-a)^2}{12}. \tag{18}$$

Univariate Continuous Random Variables

- The Gaussian distribution (or normal distribution) is undoubtedly one of the most fundamental and important distributions.
- The Gaussian distribution first appeared in 1738 during the analysis of the asymptotic behavior of binomial coefficients:
  De Moivre-Laplace Theorem: Let $X \sim \mathrm{Bin}(n, \mu)$. Then,

$$\lim_{n \to \infty} P\left(a \leq Z := \frac{X - \mathrm{E}[X]}{\sqrt{\mathrm{Var}[X]}} = \frac{X - n\mu}{\sqrt{n\mu(1-\mu)}} \leq b \right) = \int_a^b \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz. \tag{19}$$

- For a sufficiently large n, we obtain the following approximation:

$$\frac{X}{n} \approx \mu + \sqrt{\frac{\mu(1-\mu)}{n}} Z, \tag{20}$$

where Z has the PDF $\phi(z) := \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$.

Univariate Continuous Random Variables

- This motivates the definition of a class of random variables of the form $\mu + \sigma Z$, where Z is a random variable with the probability density function (PDF):

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}. \tag{21}$$

- These are called Gaussian distributions, and Z is specifically referred to as the *standard Gaussian random variable*.
- **Gaussian Random Variable**: A random variable X is said to follow the *Gaussian distribution (or normal distribution)* with parameters $\mu \in \mathbb{R}$ and $\sigma > 0$, denoted by $X \sim \mathcal{N}(\mu, \sigma^2)$, if:

$$X = \mu + \sigma Z, \tag{22}$$

for some standard Gaussian random variable $Z \sim \mathcal{N}(0, 1)$.

Univariate Continuous Random Variables

- When $X \sim \mathcal{N}(\mu, \sigma^2)$, its PDF is given by:

$$f_X(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right). \tag{23}$$

This is a popular alternative definition of the Gaussian distribution.
Proof:

$$\begin{aligned} f_X(x; \mu, \sigma^2) &= \frac{d}{dx} P(X \leq x) = \frac{d}{dx} P\left(Z \leq \frac{x-\mu}{\sigma}\right) \\ &= \frac{d}{dx} \int_{-\infty}^{(x-\mu)/\sigma} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right) dz \\ &= \frac{1}{\sigma} \left(\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)\right). \end{aligned} \tag{24}$$

Univariate Continuous Random Variables

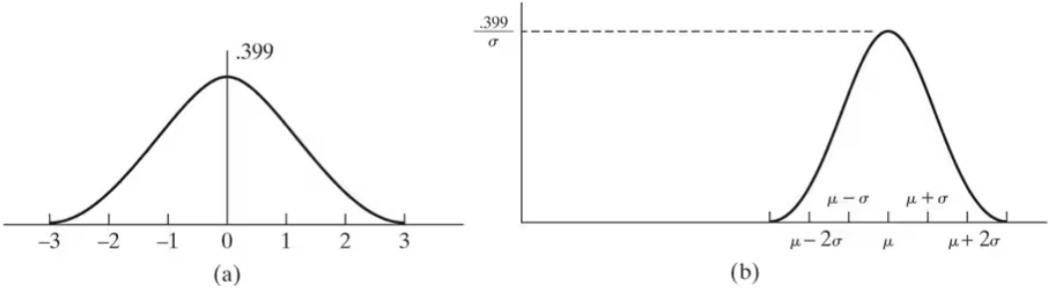


Fig. 5.5 [FCP]: Normal density function: (a) $\mu = 0, \sigma = 1$; (b) arbitrary μ, σ^2.

- We denote the PDF and CDF of the *standard Gaussian distribution* $\mathcal{N}(0, 1)$ by $\phi(z)$ and $\Phi(z)$, respectively.

Multivariate Random Variables

- **Joint (Cumulative) Distribution Function**: For a random vector $\boldsymbol{X} := (X_1, \ldots, X_d)^T$, its joint CDF is given by:

$$F_{\boldsymbol{X}}(\boldsymbol{x}) = P(X_1 \leq x_1, \ldots, X_d \leq x_d) \tag{25}$$

where $\boldsymbol{x} := (x_1, \ldots, x_d)^T \in \mathbb{R}^d$.

- **Joint Probability Mass Function**: For a jointly discrete random vector $\boldsymbol{X}$, its joint PMF $p_{\boldsymbol{X}}$ is given by:

$$p_{\boldsymbol{X}}(\boldsymbol{x}) := P(\boldsymbol{X} = \boldsymbol{x}) = P(\{s : X_1(s) = x_1, X_2(s) = x_2, \ldots, X_d(s) = x_d\}). \tag{26}$$

- The joint PMF is also related to the joint CDF. For example, when $d = 2$,

$$p(x_1, x_2) = F(x_1, x_2) - F(x_1-, x_2) - F(x_1, x_2-) + F(x_1-, x_2-). \tag{27}$$

Multivariate Random Variables

- **Example 6.1a from [FCP]**: Suppose that three balls are randomly selected from an urn containing 3 red, 4 white, and 5 blue balls. Let X and Y denote the number of red and white balls chosen, respectively. Then, the joint PMF of X and Y is given by:

$$p_{X,Y}(i,j) = \frac{\binom{3}{i}\binom{4}{j}\binom{5}{3-i-j}}{\binom{12}{3}} \tag{28}$$

for non-negative integers i and j such that $i + j \leq 3$.

Multivariate Random Variables

$i \backslash j$	0	1	2	3	Row sum $= P\{X = i\}$
0	$\frac{10}{220}$	$\frac{40}{220}$	$\frac{30}{220}$	$\frac{4}{220}$	$\frac{84}{220}$
1	$\frac{30}{220}$	$\frac{60}{220}$	$\frac{18}{220}$	0	$\frac{108}{220}$
2	$\frac{15}{220}$	$\frac{12}{220}$	0	0	$\frac{27}{220}$
3	$\frac{1}{220}$	0	0	0	$\frac{1}{220}$
Column sum $= P\{Y = j\}$	$\frac{56}{220}$	$\frac{112}{220}$	$\frac{48}{220}$	$\frac{4}{220}$	

[FCP] Tab. 6.1: $p_{X,Y}(i,j) = P(X = i, Y = j)$.

Multivariate Random Variables

- **Multinomial Random Vector**: A random vector $\boldsymbol{X}$ is said to follow the *multinomial distribution*, denoted by $\boldsymbol{X} \sim \mathrm{Multin}(n, \boldsymbol{\mu})$, where $n \in \mathbb{N}$, $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_d)^T$, $\mu_i \geq 0$, and $\sum_{i=1}^{d} \mu_i = 1$, if its PMF is given by:

$$p(\boldsymbol{x}; n, \boldsymbol{\mu}) = \begin{cases} \binom{n}{x_1, x_2, \ldots, x_d} \mu_1^{x_1} \mu_2^{x_2} \cdots \mu_d^{x_d} & \text{if } x_i \in \{0, 1, \ldots, n\} \text{ and } \sum_{i=1}^{d} x_i = n, \\ 0 & \text{otherwise.} \end{cases} \tag{29}$$

- When $\boldsymbol{X} \sim \mathrm{Multin}(n, \boldsymbol{\mu})$, the marginal distribution of the i-th coordinate is binomial:

$$X_i \sim \mathrm{Bin}(n, \mu_i). \tag{30}$$

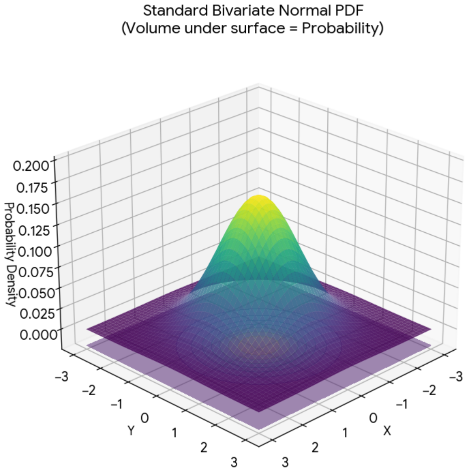
Multivariate Random Variables

- <u>**Joint Probability Density Function**</u>: For a jointly continuous random vector $\boldsymbol{X}$, its joint PDF $f_{\boldsymbol{X}}$ satisfies:
  1. $f_{\boldsymbol{X}}(\boldsymbol{x}) \geq 0$ for all $\boldsymbol{x} \in \mathbb{R}^d$.
  2. $P(\boldsymbol{X} \in (a_1, b_1) \times \cdots \times (a_d, b_d)) = \int_{a_1}^{b_1} \cdots \int_{a_d}^{b_d} f_{\boldsymbol{X}}(x_1, \ldots, x_d)\, dx_d \ldots dx_1$.
- For example, let $f_{X_1}(x_1)$ and $f_{X_2}(x_2)$ be the PDFs of X_1 and X_2, respectively. Then, the product

$$f_{X_1}(x_1) f_{X_2}(x_2), \tag{31}$$

which is a function of $(x_1, x_2)^T \in \mathbb{R}^2$, satisfies the conditions for a joint PDF.

Multivariate Random Variables



- When f_{X_1} and f_{X_2} are the PDFs of $\mathcal{N}(0, 1^2)$,

$$(x_1, x_2)^T \mapsto f_{X_1}(x_1) f_{X_2}(x_2) = \frac{1}{2\pi} \exp\left(-\frac{1}{2}\left(x_1^2 + x_2^2\right)\right) \tag{32}$$

defines a joint PDF of $\boldsymbol{X} = (X_1, X_2)^T$.

Multivariate Random Variables

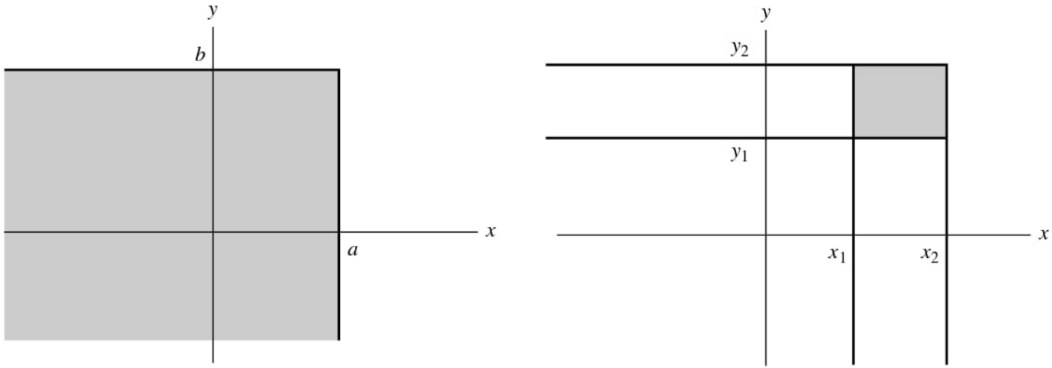
- The joint PDF is related to the joint CDF by its derivative:

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{\partial^d}{\partial x_1 \cdots \partial x_d} F_{\boldsymbol{X}}(\boldsymbol{x}). \tag{33}$$

Consequently, for a small $\epsilon_i > 0$,

$$P(x_1 - \epsilon_1/2 \leq X_1 \leq x_1 + \epsilon_1/2, \ldots, x_d - \epsilon_d/2 \leq X_d \leq x_d + \epsilon_d/2) \approx f_{\boldsymbol{X}}(\boldsymbol{x})\epsilon_1 \cdots \epsilon_d. \tag{34}$$

Multivariate Random Variables



Figs. 3.1 and 3.2 [MSDA]:
$P(x_1 < X \leq x_2, y_1 < Y \leq y_2) = F(x_2, y_2) - F(x_2, y_1) - F(x_1, y_2) + F(x_1, y_1)$

Multivariate Random Variables

- Given $p_{X,Y}(x, y)$, we can calculate:

$$P(X = x) = P(X = x, Y \in \mathcal{Y}) = \sum_{y \in \mathcal{Y}} P(X = x, Y = y) = \sum_{y \in \mathcal{Y}} p_{X,Y}(x, y). \tag{35}$$

- **Marginal Probability Mass Function**: Let $\boldsymbol{X}$ and $\boldsymbol{Y}$ be random vectors with joint PMF $p_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x}, \boldsymbol{y})$. Then, the *marginal PMF* $p_{\boldsymbol{X}}(\boldsymbol{x})$ is given by:

$$p_{\boldsymbol{X}}(\boldsymbol{x}) = \sum_{\boldsymbol{y} \in \mathcal{Y}} p_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x}, \boldsymbol{y}) \tag{36}$$

Multivariate Random Variables

- Given the joint PDF $f_{X,Y}(x, y)$, we can derive the marginal PDF as follows:

$$
\begin{aligned}
f_X(x) := \lim_{\epsilon \to 0^+} \frac{P(x - \epsilon/2 \leq X \leq x + \epsilon/2)}{\epsilon} &= \lim_{\epsilon \to 0^+} \frac{1}{\epsilon} \int_{-\infty}^{\infty} \int_{x-\epsilon/2}^{x+\epsilon/2} f_{X,Y}(u, y)\, du\, dy \\
&= \int_{-\infty}^{\infty} \left(\lim_{\epsilon \to 0^+} \frac{1}{\epsilon} \int_{x-\epsilon/2}^{x+\epsilon/2} f_{X,Y}(u, y)\, du \right) dy \\
&= \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dy.
\end{aligned} \tag{37}
$$

- **Marginal Probability Density Function**: Let $\boldsymbol{X}$ and $\boldsymbol{Y}$ be random vectors with joint PDF $f_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x}, \boldsymbol{y})$. Then, the *marginal PDF* of $\boldsymbol{X}$, denoted $f_{\boldsymbol{X}}(\boldsymbol{x})$, is given by:

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \int_{\mathbb{R}^{d_{\boldsymbol{Y}}}} f_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x}, \boldsymbol{y})\, d\boldsymbol{y}. \tag{38}$$

Outline

Means and Covariances

- For a multivariate r.v. $\boldsymbol{X}(\cdot) := (X_1(\cdot), \ldots, X_d(\cdot))^T$, its expectation is defined as:

$$\boldsymbol{\mu}_{\boldsymbol{X}} = \mathrm{E}_{\boldsymbol{X}}[\boldsymbol{X}] = \left(\mathrm{E}_{X_1}[X_1], \ldots, \mathrm{E}_{X_d}[X_d]\right)^T \in \mathbb{R}^d. \tag{39}$$

- The expectation operator is linear. For any real constants a and b, and any real-valued functions g and h defined on the support of $\boldsymbol{X}$,

$$\mathrm{E}[ag(\boldsymbol{X}) + bh(\boldsymbol{X})] = a\,\mathrm{E}[g(\boldsymbol{X})] + b\,\mathrm{E}[h(\boldsymbol{X})]. \tag{40}$$

Means and Covariances

- **Covariance**: The *covariance* between two univariate r.v.s X and Y is given by:

$$\sigma_{X,Y} = \mathrm{Cov}_{X,Y}[X,Y] = \mathrm{E}_{X,Y}\left[(X - \mathrm{E}_X[X])(Y - \mathrm{E}_Y[Y])\right]. \tag{41}$$

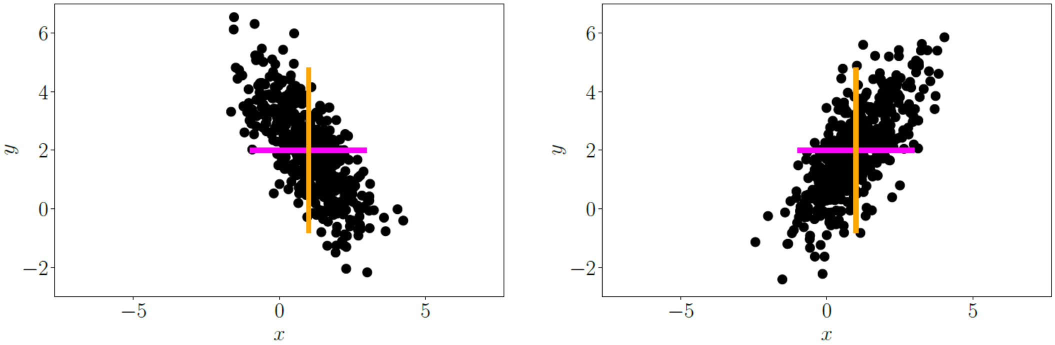
- For a $d_{\boldsymbol{X}}$-dimensional $\boldsymbol{X}$ and a $d_{\boldsymbol{Y}}$-dimensional $\boldsymbol{Y}$, their covariance is defined as:

$$\Sigma_{\boldsymbol{X},\boldsymbol{Y}} = \mathrm{Cov}_{\boldsymbol{X},\boldsymbol{Y}}[\boldsymbol{X},\boldsymbol{Y}] = \mathrm{E}_{\boldsymbol{X},\boldsymbol{Y}}[(\boldsymbol{X} - \mathrm{E}_{\boldsymbol{X}}[\boldsymbol{X}])(\boldsymbol{Y} - \mathrm{E}_{\boldsymbol{Y}}[\boldsymbol{Y}])^T] \in \mathbb{R}^{d_{\boldsymbol{X}} \times d_{\boldsymbol{Y}}}. \tag{42}$$

- A useful shortcut formula is:

$$\mathrm{Cov}[\boldsymbol{X},\boldsymbol{Y}] = \mathrm{E}[\boldsymbol{X}\boldsymbol{Y}^T] - \mathrm{E}[\boldsymbol{X}]\,\mathrm{E}[\boldsymbol{Y}]^T. \tag{43}$$

Means and Covariances



[MML] Fig. 6.5: Scatter plots of two observed random variables X and Y.

Means and Covariances

- **Variance**: The *variance* of a r.v. $\boldsymbol{X}$ is defined as:

$$\Sigma_{\boldsymbol{X}} = \mathrm{Var}_{\boldsymbol{X}}[\boldsymbol{X}] = \mathrm{E}_{\boldsymbol{X}}[(\boldsymbol{X} - \mathrm{E}_{\boldsymbol{X}}[\boldsymbol{X}])(\boldsymbol{X} - \mathrm{E}_{\boldsymbol{X}}[\boldsymbol{X}])^T] \tag{44}$$

  This matrix is referred to as the *covariance matrix* of $\boldsymbol{X}$.
- Since $\mathrm{Var}[\boldsymbol{X}] = \mathrm{Cov}[\boldsymbol{X}, \boldsymbol{X}]$, based on the shortcut formula of covariance,

$$\mathrm{Var}[\boldsymbol{X}] = \mathrm{E}[\boldsymbol{X}\boldsymbol{X}^T] - \mathrm{E}[\boldsymbol{X}]\,\mathrm{E}[\boldsymbol{X}^T]. \tag{45}$$

- For any functions g and h defined on the support of $\boldsymbol{X}$ and $\boldsymbol{Y}$, respectively,

$$\mathrm{Cov}[g(\boldsymbol{X}), h(\boldsymbol{Y})] = \mathrm{E}\left[\left(g(\boldsymbol{X}) - \mathrm{E}[g(\boldsymbol{X})]\right)\left(h(\boldsymbol{Y}) - \mathrm{E}[h(\boldsymbol{Y})]\right)^T\right]. \tag{46}$$

Means and Covariances

- **Correlation**: Let X and Y be univariate random variables having positive variances. The *correlation* between X and Y is given by:

$$\rho_{X,Y} = \mathrm{Corr}_{X,Y}[X, Y] = \frac{\mathrm{Cov}_{X,Y}[X, Y]}{\sqrt{\mathrm{Var}_X[X]\,\mathrm{Var}_Y[Y]}}. \tag{47}$$

- By the Cauchy-Schwartz inequality, $-1 \leq \mathrm{Corr}(X, Y) \leq 1$ always holds.
- Again, the *standardized variable* (or Z*-score*) of X is defined as:

$$Z_X = \frac{X - \mathrm{E}[X]}{\sqrt{\mathrm{Var}[X]}}. \tag{48}$$

The correlation is equal to the covariance between Z-scores:

$$\mathrm{Corr}[X, Y] = \mathrm{Cov}\left[\frac{X - \mathrm{E}[X]}{\sqrt{\mathrm{Var}[X]}}, \frac{Y - \mathrm{E}[Y]}{\sqrt{\mathrm{Var}[Y]}}\right]. \tag{49}$$

Means and Covariances

- **Sample Mean (Empirical Mean)**: Let $x_1, \ldots, x_n$ be n observations of a random variable X. Then, the *sample mean* of X is defined as $\bar{x} = n^{-1} \sum_{i=1}^{n} x_i$.
- **Sample Covariance**: Let $(x_1, y_1)^T, \ldots, (x_n, y_n)^T$ be n observations of a random vector $(X, Y)^T$. Then, the *sample covariance* between X and Y is defined as $s_{X,Y} = (n-1)^{-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$.
- **Sample Variance**: Let $x_1, \ldots, x_n$ be n observations of a random variable X. Then, the *sample variance* of X is defined as $s_{X,X} = s_X^2 = (n-1)^{-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$.
- **Sample Correlation**: Let $(x_1, y_1)^T, \ldots, (x_n, y_n)^T$ be n observations of a random vector $(X, Y)^T$. Assume $s_X > 0$ and $s_Y > 0$. Then, the *sample correlation* is defined as $r_{X,Y} = s_{X,Y}/(s_X s_Y)$.
- We can similarly define multivariate versions of the sample mean and covariance matrix.

Means and Covariances

- For any two random vectors $\boldsymbol{X} \in \mathbb{R}^d$ and $\boldsymbol{Y} \in \mathbb{R}^d$:
  - $\mathrm{E}[\boldsymbol{X} + \boldsymbol{Y}] = \mathrm{E}[\boldsymbol{X}] + \mathrm{E}[\boldsymbol{Y}]$.
  - $\mathrm{E}[\boldsymbol{X} - \boldsymbol{Y}] = \mathrm{E}[\boldsymbol{X}] - \mathrm{E}[\boldsymbol{Y}]$.
  - $\mathrm{Var}[\boldsymbol{X} + \boldsymbol{Y}] = \mathrm{Var}[\boldsymbol{X}] + \mathrm{Var}[\boldsymbol{Y}] + \mathrm{Cov}[\boldsymbol{X}, \boldsymbol{Y}] + \mathrm{Cov}[\boldsymbol{X}, \boldsymbol{Y}]^T$.
  - $\mathrm{Var}[\boldsymbol{X} - \boldsymbol{Y}] = \mathrm{Var}[\boldsymbol{X}] + \mathrm{Var}[\boldsymbol{Y}] - \mathrm{Cov}[\boldsymbol{X}, \boldsymbol{Y}] - \mathrm{Cov}[\boldsymbol{X}, \boldsymbol{Y}]^T$.
- Let $\boldsymbol{X} \in \mathbb{R}^{d_{\boldsymbol{X}}}$ and $\boldsymbol{Y} \in \mathbb{R}^{d_{\boldsymbol{Y}}}$ be random vectors. For any constant matrices A and B, and constant vectors $\boldsymbol{b}, \boldsymbol{c}, \boldsymbol{d}$ of appropriate dimensions:
  - $\mathrm{E}[A\boldsymbol{X} + \boldsymbol{b}] = A\,\mathrm{E}[\boldsymbol{X}] + \boldsymbol{b}$.
  - $\mathrm{Var}[A\boldsymbol{X} + \boldsymbol{b}] = A\,\mathrm{Var}[\boldsymbol{X}]A^T$.
  - $\mathrm{Cov}[A\boldsymbol{X} + \boldsymbol{c}, B\boldsymbol{Y} + \boldsymbol{d}] = A\mathrm{Cov}[\boldsymbol{X}, \boldsymbol{Y}]B^T$.

Statistical Independence

- **Statistical Independence**: Two random variables X and Y are said to be *independent* if, for any two (measurable) sets T_X and T_Y in their target spaces,

$$P(X \in T_X, Y \in T_Y) = P(X \in T_X)P(Y \in T_Y). \tag{50}$$

  That is, the events $X^{-1}(T_X)$ and $Y^{-1}(T_Y)$ are independent.
- We write $X \perp\!\!\!\perp Y$ to denote that X is independent of Y.
- When the target spaces are real coordinate spaces, two random vectors $\boldsymbol{X}$ and $\boldsymbol{Y}$ are independent if and only if either of the following equivalent conditions holds:
  1. $F_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x}, \boldsymbol{y}) = F_{\boldsymbol{X}}(\boldsymbol{x})F_{\boldsymbol{Y}}(\boldsymbol{y})$
  2. $f_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x}, \boldsymbol{y}) = f_{\boldsymbol{X}}(\boldsymbol{x})f_{\boldsymbol{Y}}(\boldsymbol{y})$ (or $p_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x}, \boldsymbol{y}) = p_{\boldsymbol{X}}(\boldsymbol{x})p_{\boldsymbol{Y}}(\boldsymbol{y})$ in the discrete case).
- For any (measurable) functions g and h, if $X \perp\!\!\!\perp Y$, then $g(X) \perp\!\!\!\perp h(Y)$.

Statistical Independence

- The random vectors $\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_n$ are said to be independent if, for all measurable sets $T_1, T_2, \ldots, T_n$ in their respective target spaces,

$$P(\boldsymbol{X}_1 \in T_1, \boldsymbol{X}_2 \in T_2, \ldots, \boldsymbol{X}_n \in T_n) = \prod_{i=1}^{n} P(\boldsymbol{X}_i \in T_i). \tag{51}$$

- When the target spaces are real coordinate spaces, these random vectors are independent if and only if the following equivalent conditions hold:
  1. $F_{\boldsymbol{X}_1,\ldots,\boldsymbol{X}_n}(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n) = F_{\boldsymbol{X}_1}(\boldsymbol{x}_1) \cdots F_{\boldsymbol{X}_n}(\boldsymbol{x}_n)$
  2. $f_{\boldsymbol{X}_1,\ldots,\boldsymbol{X}_n}(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n) = f_{\boldsymbol{X}_1}(\boldsymbol{x}_1) \cdots f_{\boldsymbol{X}_n}(\boldsymbol{x}_n)$ (or $p_{\boldsymbol{X}_1,\ldots,\boldsymbol{X}_n}(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n) = p_{\boldsymbol{X}_1}(\boldsymbol{x}_1) \cdots p_{\boldsymbol{X}_n}(\boldsymbol{x}_n)$ in the discrete case).
- An infinite collection of random vectors is said to be independent if every finite sub-collection is independent.

Statistical Independence

- For any two independent random vectors $\boldsymbol{X} \in \mathbb{R}^d$ and $\boldsymbol{Y} \in \mathbb{R}^d$:

$$\mathrm{Cov}[\boldsymbol{X}, \boldsymbol{Y}] = 0, \tag{52}$$

implying $\mathrm{Var}[\boldsymbol{X} + \boldsymbol{Y}] = \mathrm{Var}[\boldsymbol{X}] + \mathrm{Var}[\boldsymbol{Y}]$. However, the converse does not hold in general.

- **Example 6.5 from [MML]**: Consider a random variable X with $\mathrm{E}[X] = 0$ and $\mathrm{E}[X^3] = 0$. Let $Y = X^2$. Then, $\mathrm{Cov}[X, Y] = 0$. Do you think Y is independent of X? That is, if we observe X, does this observation truly provide no information regarding Y?

Outline

Multivariate Gaussian Distributions

- We first define the *multivariate standard Gaussian random vector* $\boldsymbol{Z}$:

$$f_{\boldsymbol{Z}}(\boldsymbol{z}) = \prod_{i=1}^{d} \phi(z_i) = (2\pi)^{-d/2} \exp\left(-\boldsymbol{z}^T \boldsymbol{z}/2\right), \quad \text{denoted by } \boldsymbol{Z} \sim \mathcal{N}(0, I_d). \tag{53}$$

- The multivariate Central Limit Theorem states that the sample mean vector is approximately distributed as:

$$\frac{\boldsymbol{X}_1 + \cdots + \boldsymbol{X}_n}{n} \approx \boldsymbol{\mu} + \Sigma^{1/2} \boldsymbol{Z}, \tag{54}$$

for some $\boldsymbol{\mu} \in \mathbb{R}^d$, $\Sigma^{1/2} \in \mathbb{R}^{d \times d}$, and $\boldsymbol{Z} \sim \mathcal{N}(0, I_d)$, where $\Sigma^{1/2}$ is symmetric and $\Sigma^{1/2}\Sigma^{1/2} := \Sigma$ is symmetric and positive definite.

Multivariate Gaussian Distributions

- **Multivariate Gaussian Distribution**: A random vector $\boldsymbol{X} \in \mathbb{R}^d$ is said to follow the *multivariate Gaussian distribution* with mean $\boldsymbol{\mu} \in \mathbb{R}^d$ and symmetric and positive definite matrix $\Sigma \in \mathbb{R}^{d \times d}$, denoted by $\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, if:

$$\boldsymbol{X} = \boldsymbol{\mu} + \Sigma^{1/2}\boldsymbol{Z}, \quad \text{where } \boldsymbol{Z} \sim \mathcal{N}(0, I_d). \tag{55}$$

  We also use the notation $\mathcal{N}(\boldsymbol{x}; \boldsymbol{\mu}, \Sigma)$ to refer to the multivariate Gaussian PDF.
- To derive the joint PDF of $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$, we first derive the change of variables formula, which describes how a probability density transforms under a one-to-one differentiable mapping.

Multivariate Gaussian Distributions

- Let $\boldsymbol{X}$ be a continuous random vector with PDF $f_{\boldsymbol{X}}(\boldsymbol{x})$ supported on a region $T_X \subseteq \mathbb{R}^d$. Let $\boldsymbol{Y} = g(\boldsymbol{X})$, where g is a one-to-one, differentiable transformation from T_X onto T_Y. Then, $\boldsymbol{Y}$ has the PDF:

$$f_{\boldsymbol{Y}}(\boldsymbol{y}) = \begin{cases} f_{\boldsymbol{X}}(g^{-1}(\boldsymbol{y})) \left|\det \frac{\partial g^{-1}(\boldsymbol{y})}{\partial \boldsymbol{y}}\right|, & \text{if } \boldsymbol{y} \in T_Y, \\ 0, & \text{otherwise.} \end{cases} \tag{56}$$

Here, $\left|\det \frac{\partial g^{-1}(\boldsymbol{y})}{\partial \boldsymbol{y}}\right|$ denotes the absolute value of the determinant of the Jacobian matrix.

Multivariate Gaussian Distributions

- When $g : \mathbb{R}^d \to \mathbb{R}^d$ is differentiable and invertible on T_X, and $g^{-1}(\boldsymbol{y}) \in T_X$, denoting the hypercube $C_\epsilon(\boldsymbol{y}) = [\boldsymbol{y} - \frac{\epsilon}{2}\mathbf{1}, \boldsymbol{y} + \frac{\epsilon}{2}\mathbf{1}]$ where $\mathbf{1} := (1, \ldots, 1)^T$:

$$
\begin{aligned}
f_{\boldsymbol{Y}}(\boldsymbol{y}) &= \lim_{\epsilon \to 0^+} \frac{P(\boldsymbol{Y} \in C_\epsilon(\boldsymbol{y}))}{\epsilon^d} \\
&= \lim_{\epsilon \to 0^+} \frac{P(\boldsymbol{X} \in g^{-1}(C_\epsilon(\boldsymbol{y})))}{\epsilon^d} \\
&= \lim_{\epsilon \to 0^+} \frac{P(\boldsymbol{X} \in g^{-1}(C_\epsilon(\boldsymbol{y})))}{\mathrm{Vol}(g^{-1}(C_\epsilon(\boldsymbol{y})))} \frac{\mathrm{Vol}(g^{-1}(C_\epsilon(\boldsymbol{y})))}{\epsilon^d} \\
&= f_{\boldsymbol{X}}(g^{-1}(\boldsymbol{y})) \left| \det \frac{\partial g^{-1}(\boldsymbol{y})}{\partial \boldsymbol{y}} \right| .
\end{aligned}
\tag{57}
$$

Multivariate Gaussian Distributions

- **Change of Variables Formula**: Suppose there exists a partition of T_X such that $T_X = \bigcup_{i=1}^{k} T_{X,i}$ (where $T_{X,i} \cap T_{X,j} = \emptyset$ for $i \neq j$), and the restriction of g to $T_{X,i}$, denoted by g_i, is a one-to-one differentiable mapping. Then, $\boldsymbol{Y}$ has the PDF:

$$f_{\boldsymbol{Y}}(\boldsymbol{y}) = \sum_{i=1}^{k} f_{\boldsymbol{X}}(g_i^{-1}(\boldsymbol{y})) \left| \det \frac{\partial g_i^{-1}(\boldsymbol{y})}{\partial \boldsymbol{y}} \right|, \tag{58}$$

  where the sum is over indices i such that $\boldsymbol{y}$ is in the image of g_i.
- The change of variables formula is fundamental in defining multivariate distributions.

Multivariate Gaussian Distributions

- Let $\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$. That is, $\boldsymbol{X} = \boldsymbol{\mu} + \Sigma^{1/2}\boldsymbol{Z}$ where $\boldsymbol{Z} \sim \mathcal{N}(0, I_d)$. Then, the PDF of $\boldsymbol{X}$ is given by:

$$f_{\boldsymbol{X}}(\boldsymbol{x}; \boldsymbol{\mu}, \Sigma) = (2\pi)^{-d/2}|\Sigma|^{-1/2} \exp\left(-\frac{1}{2}(\boldsymbol{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\boldsymbol{x} - \boldsymbol{\mu})\right). \tag{59}$$

Sketch of Proof: Note that $\boldsymbol{Z} = \Sigma^{-1/2}(\boldsymbol{X} - \boldsymbol{\mu})$ implies that $g(\boldsymbol{z}) = \boldsymbol{\mu} + \Sigma^{1/2}\boldsymbol{z}$ is a one-to-one differentiable mapping. By the change of variables formula,

$$f_{\boldsymbol{X}}(\boldsymbol{x}; \boldsymbol{\mu}, \Sigma) = f_{\boldsymbol{Z}}(g^{-1}(\boldsymbol{x})) \left| \det \frac{\partial g^{-1}(\boldsymbol{x})}{\partial \boldsymbol{x}} \right|. \tag{60}$$

Multivariate Gaussian Distributions

- For example, for any bivariate Gaussian random vector

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix} \right), \tag{61}$$

its PDF is given by:

$$f_{X,Y}(x, y; \boldsymbol{\mu}, \Sigma) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)} \left(\left(\frac{x-\mu_X}{\sigma_X} \right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y} \right)^2 \right.\right. \\ \left.\left. -2\rho \left(\frac{x-\mu_X}{\sigma_X} \right) \left(\frac{y-\mu_Y}{\sigma_Y} \right) \right) \right]. \tag{62}$$

Multivariate Gaussian Distributions

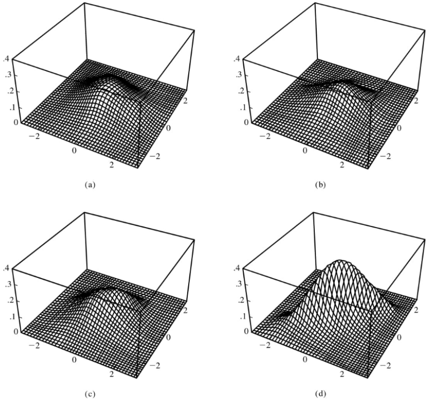


Fig. 3.9. [MSDA]

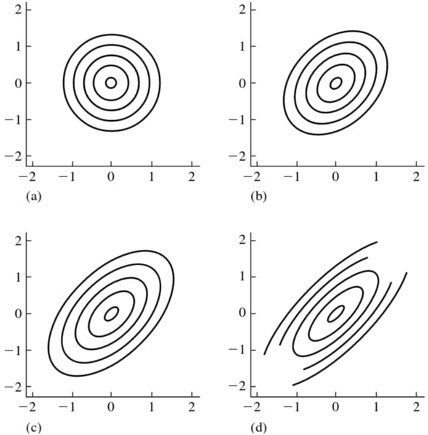


Fig. 3.10. [MSDA]

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